Warehouse Inventory Reconciliation

Problem statement

Compare opening stock, incoming quantities, dispatched quantities, and physical closing stock for multiple products; identify discrepancies and calculate replenishment quantities.

Assumptions and rules

• Each product record contains name, opening stock, incoming stock, dispatched stock, and physical closing stock.

• Expected closing stock = opening + incoming - dispatched.

• Discrepancy = physical closing - expected closing.

• A product is replenished up to its target stock level when physical stock is below that target.

Algorithm

1. Read product records and target stock levels.

2. For each product, unpack the record.

3. Calculate expected closing stock.

4. Compare expected and physical closing stock to find discrepancy.

5. If physical stock is below target, calculate replenishment quantity.

6. Print a reconciliation report for every product.

Pseudocode

```
START\nINPUT inventory records\nFOR each product record\n  UNPACK name, opening, incoming, dispatched, physical\n  expected
```

Flowchart

```
[Start] -> [Read inventory records]\n   |\n[Select next product]\n   |\n[Calculate expected closing stock]\n   |\n[Compare
```

Python solution

The program uses beginner-level Python constructs from the assigned syllabus: variables, data types, operators, conditions, loops, strings, lists, tuples, and unpacking where appropriate.

```
inventory = [
    ("Rice", 100, 50, 80, 68, 100),
    ("Wheat", 80, 20, 40, 60, 75),
    ("Sugar", 60, 30, 25, 65, 80)
]
# Record: name, opening, incoming, dispatched, physical closing, target

print("WAREHOUSE RECONCILIATION")
for item in inventory:
    name, opening, incoming, dispatched, physical, target = item

    expected = opening + incoming - dispatched
    discrepancy = physical - expected

    if discrepancy == 0:
        status = "MATCH"
    elif discrepancy > 0:
        status = "SURPLUS"
    else:
        status = "SHORTAGE"

    if physical < target:
        replenishment = target - physical
    else:
        replenishment = 0

    print(name)
```

```
    print("Expected closing:", expected)
    print("Physical closing:", physical)
    print("Discrepancy:", discrepancy, status)
    print("Replenishment required:", replenishment)
    print("-------------------------")
```

Sample output

```
Rice\nExpected closing: 70\nPhysical closing: 68\nDiscrepancy: -2 SHORTAGE\nReplenishment required: 32\n-------------------
```

Explanation / observations

A negative discrepancy indicates fewer physical units than expected. The target stock is a simple fixed reorder level for this exercise.

Conclusion

This solution models the stated case using a clear sequence of input, processing, decisions, repetition, and output. The assumptions and sample data can be changed to test other cases.